

Introduction to AI&ML

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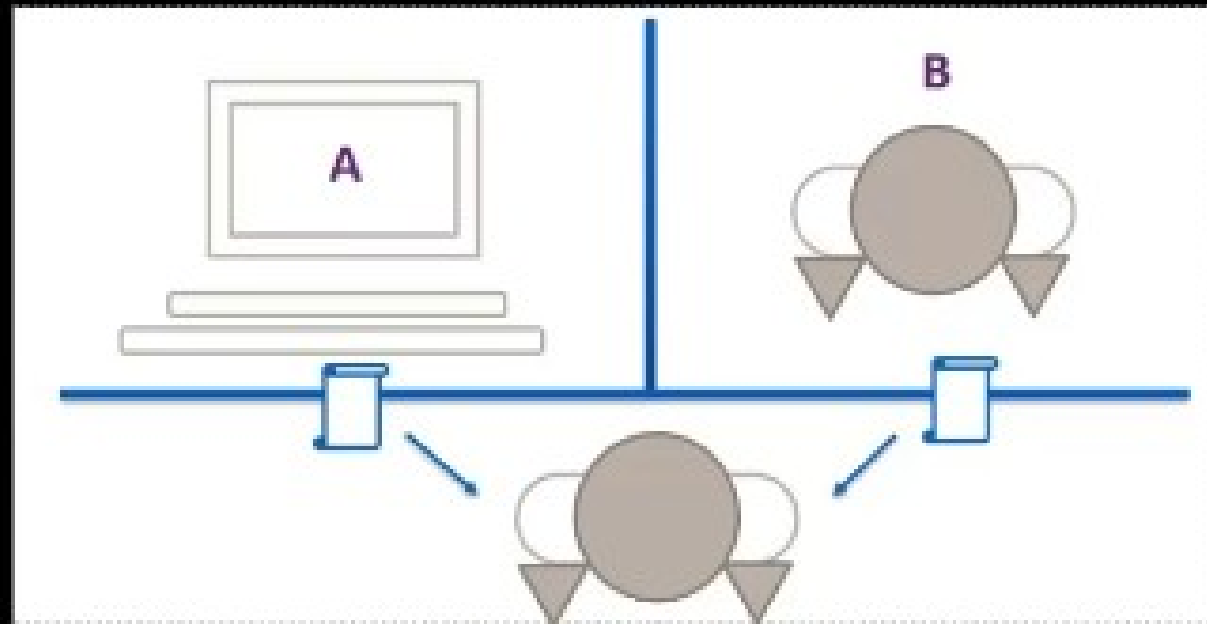
What is AI?

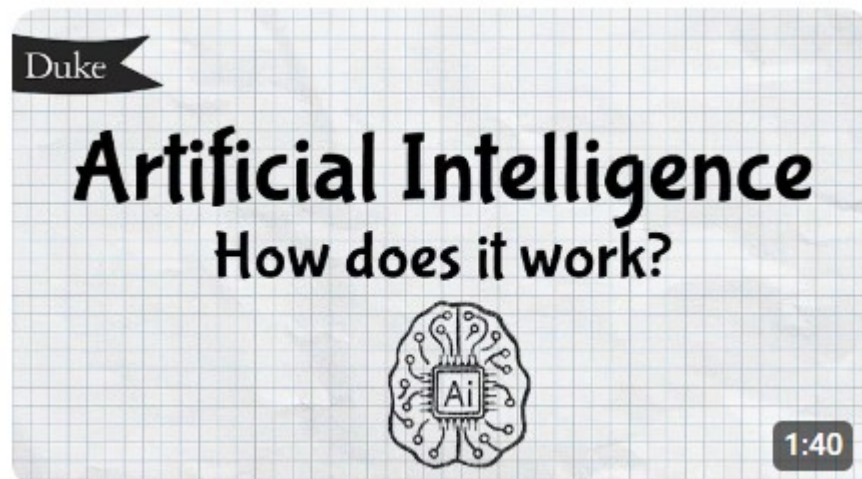
Artificial intelligence (AI) refers to computer systems capable of performing complex tasks that historically only a human could do, such as reasoning, making decisions, or solving problems.

Can you think of a situation where AI might struggle to perform as well as a human? Why?

Turing Test

Can a computer trick a human into thinking that it is human over text messages?





What is Artificial Intelligence? | Quick Learner

264K views • 3 years ago



Duke University ✓

Science fiction has certainly familiarized the world with the concept of **artificial intelligence**. But outside of Hollywood...wh...

4K CC

<https://youtu.be/c0m6yaGlZh4?si=pflpv8gqeLDqFW9j>



Coca-Cola® Masterpiece

3.6M views • 2 years ago



Coca-Cola ✓

See what happens when the whole universe comes together in Real Magic ways to help a boy in need of uplift and a little ...

https://youtu.be/VGa1imApfdg?si=97XbgexCRywWRzC_

Types of AI

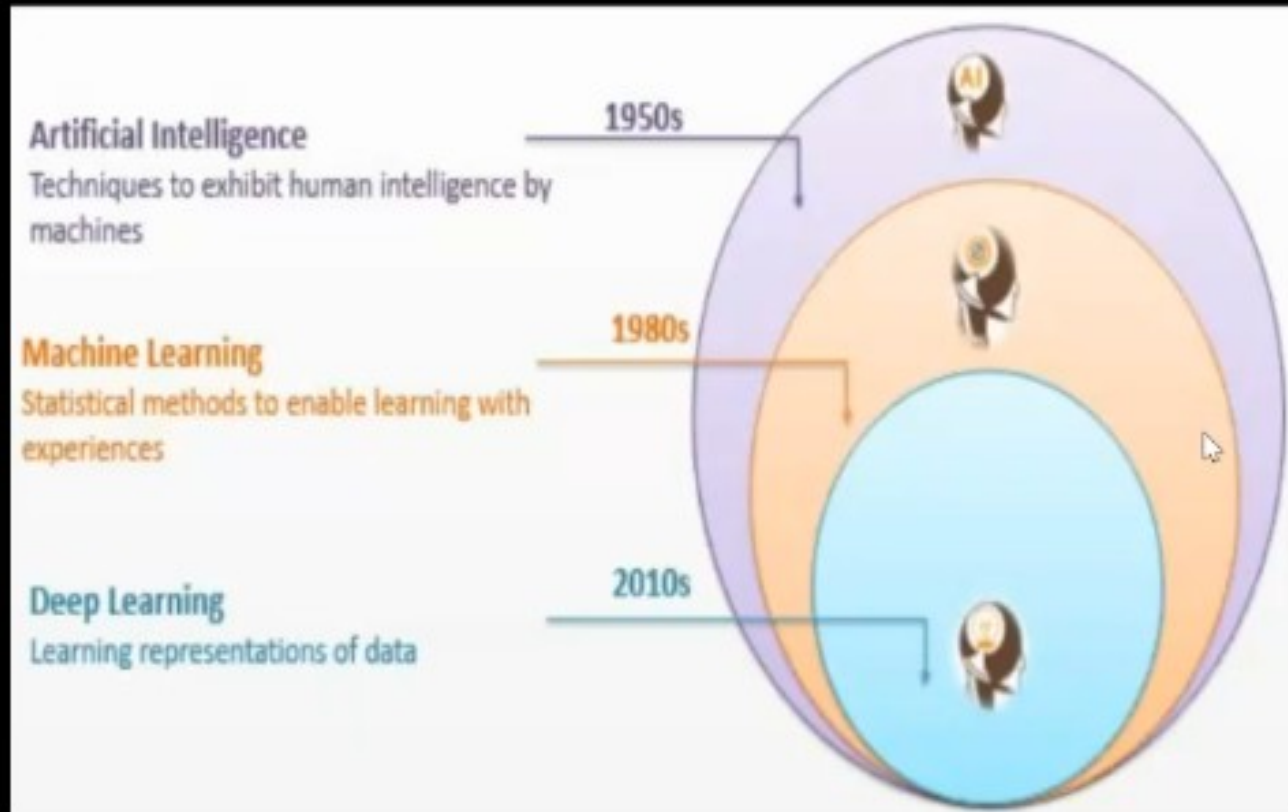
- 1. Narrow or Weak AI:** Designed to perform a narrow task
Example: Virtual assistants like Siri, Alexa, or Google Assistant.
What it does: Performs a specific task, like answering questions or setting reminders.
- 2. General or Strong AI:** Has the ability to perform any intellectual task that a human being can do. It's more theoretical and doesn't exist yet.
Example: A hypothetical AI that can learn and apply knowledge like a human.
What it does: Can perform any intellectual task that a human can.
- 3. Superintelligence:** An AI that surpasses human intelligence in all aspects. This is a theoretical concept.
Example: A hypothetical AI that surpasses human intelligence in all domains.
What it does: Can solve complex problems that humans can't.

Which of the following is AI generated Image?



What is Machine Learning?

A subset of AI, which deals with all the statistical methods to improve the machine's understanding of the data.



Types of Learning



Supervised Learning



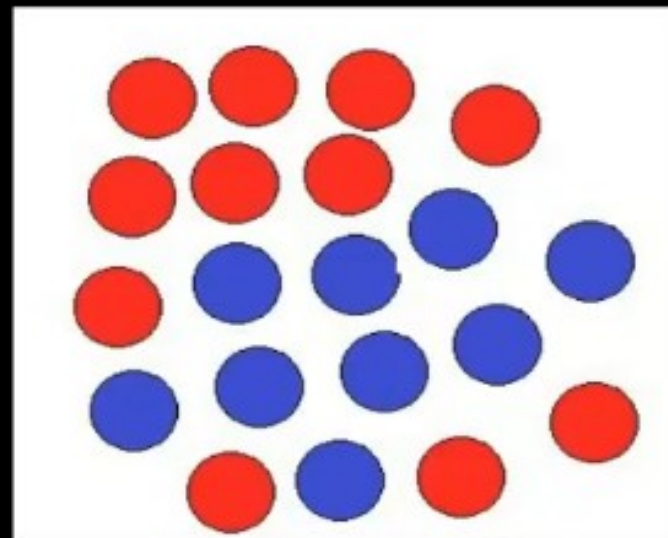
Unsupervised learning

Problem 1

We have provided some balls and two buckets.

You are asked to place/arrange the balls in two buckets.

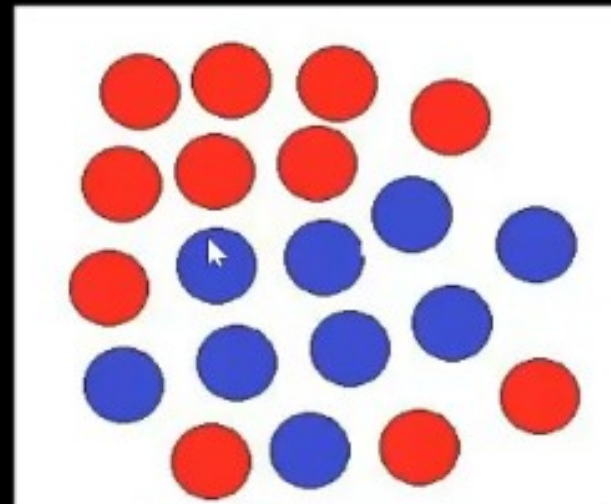
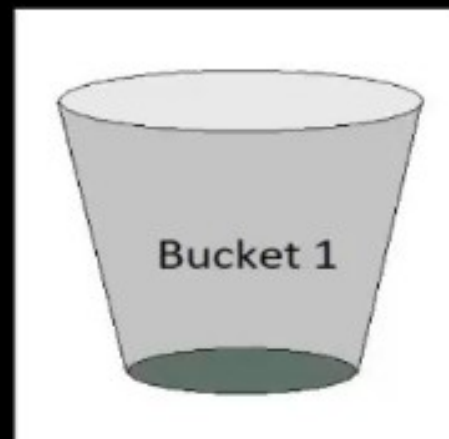
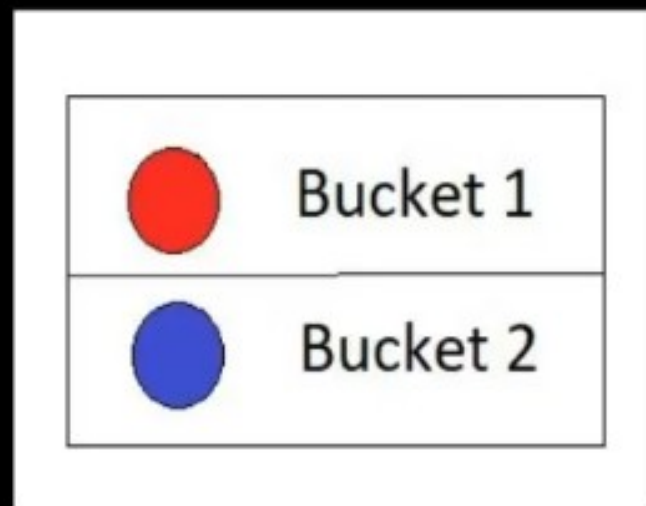
- What are the possible ways of arranging the balls?



Building on belief

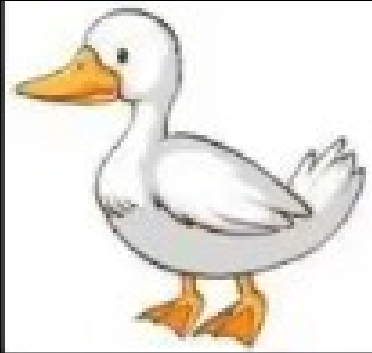
Problem 2

Arrange the two-colored balls in two buckets.



Understanding

- **Unsupervised Learning** : Involves no prior knowledge about trends or patterns in data. (problem 1)
- **Supervised Learning** : Deals with labelled data. Tries to predict an unknown input by using the past experiences. (problem 2)



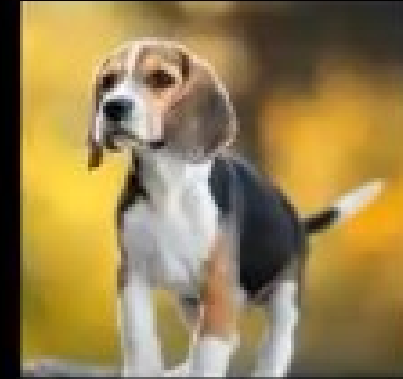
Dinosaur



Dinosaur



Elephant



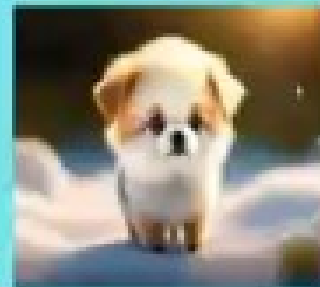
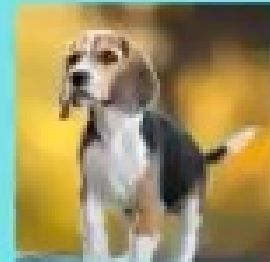
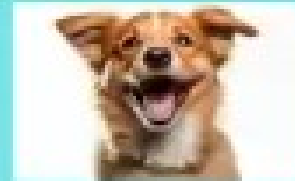
Elephant



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Building on belief

Dataset

Find the gender

Applicant Id	Name	Date of Birth	State name	Height	Weight	Gender
1	Praharsh Dixit	02/09/1993	Madhya Pradesh	178.0	69.0	M
2	Sangeeta Gorai	04/20/1993	West bengal	NaN	NaN	F
3	Sreeparvathy.S	01/07/1993	Kerala	161.0	63.8	F
4	Akash	04/04/1994	Delhi	174.0	62.0	M
5	Midhat Fatima Rizvi	06/21/1994	Uttar Pradesh	154.0	56.1	F
...	...	...	...	...	...	...
1514	Remya Ravi	10/28/1994	Harayana	151.0	61.8	F
1515	Ajit Yadav	04/30/1993	Uttaranchal	186.0	78.0	M
1516	Ankit	04/17/1994	Assam	172.0	52.0	M
1517	Shashank Sharma	10/27/1994	Uttaranchal	174.0	51.0	M
1518	Nipam Baishya	08/21/1993	Assam	170.0	76.0	M



Features



Target



Data point



Feature matrix

Continuous Vs Discrete

- Continuous Data : The number of values are infinitely countable.
- Discrete Data : rigidly defined, countable set of values

Types of problems

- Unsupervised Learning
 - Clustering problem
- Supervised Learning
 - Regression
 - Classification

Types of Supervised Learning

A. Classification

•**Definition:** Classification models predict **discrete (categorical) labels** for given input data.

•**Example:** Identifying whether an email is **spam or not spam**.

•**Algorithms:**

- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)
- Neural Networks

B. Regression

•**Definition:** Regression models predict **continuous numerical values** rather than categories.

•**Example:** Predicting **house prices** based on features like area, number of rooms, and location.

•**Algorithms:**

- Linear Regression
- Polynomial Regression
- Ridge & Lasso Regression
- Support Vector Regression (SVR)

C. Time Series Forecasting

•**Definition:** This is used for predicting values over time. The model learns from past time-stamped data and forecasts future values.

•**Example:** Stock market price prediction, weather forecasting.

•**Algorithms:**

- ARIMA (Auto-Regressive Integrated Moving Average)
- LSTM (Long Short-Term Memory)
- Prophet (by Facebook)

Types of Unsupervised Learning

A. Clustering

•**Definition:** Clustering groups similar data points together based on patterns.

•**Example:** Customer segmentation for marketing strategies.

•**Algorithms:**

- K-Means
- Hierarchical Clustering
- DBSCAN (Density-Based Spatial Clustering)

B. Dimensionality Reduction

•**Definition:** Reducing the number of features in a dataset while retaining meaningful information.

•**Example:** Image compression, feature extraction for ML models.

•**Algorithms:**

- Principal Component Analysis (PCA)
- t-SNE (t-Distributed Stochastic Neighbor Embedding)
- Autoencoders

C. Anomaly Detection

•**Definition:** Identifies rare or abnormal patterns in data.

•**Example:** Fraud detection in banking transactions.

•**Algorithms:**

- Isolation Forest
- One-Class SVM
- Local Outlier Factor (LOF)

D. Association Rule Learning

•**Definition:** Finds relationships between variables in large datasets.

•**Example:** Market basket analysis (e.g., people who buy bread often buy butter).

•**Algorithms:**

- Apriori Algorithm
- FP-Growth

Machine Learning Life Cycle

Problem statement

Data selection

Descriptive statistics

EDA

Data Pre-processing

Data Transformation

Feature selection

Model selection

Model Training

Model Evaluation

Thank you